

## COONABARABRAN AP, Australia

WMO: 957280

Lat: 31.3303S

Lon: 149.2700E

Elev: 646

StdP: 93.81

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 2.8        | 3.7 | -7.5                          | 2.2 | 17.4 | -5.3 | 2.6 | 16.7 | 9.5                   | 11.5 | 8.9 | 10.4 | 4.6                   | 210  | 0.498 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 11.5                   | 35.1            | 18.8 | 33.4 | 18.2 | 31.7 | 17.9 | 21.4                | 28.4 | 20.6 | 27.4 | 20.0 | 26.7 | 5.3                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.7                            | 15.6 | 22.5 | 18.9 | 14.8 | 22.5 | 18.1 | 14.0 | 22.0 | 65.4          | 28.0 | 62.6 | 27.3 | 60.2 | 26.6 | 24.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.5           | 8.5  | 7.8 | 0.6                        | 38.0 | 0.7                | 2.6 | 0.2                                                | 39.9 | -0.3       | 41.4 | -0.6       | 42.8 | -1.1       | 44.7 |    |    |
| (5)               |      |     | -0.2                       | 22.9 | 0.5                | 0.9 | -0.6                                               | 23.5 | -0.9       | 24.0 | -1.1       | 24.5 | -1.5       | 25.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.9   | 24.2  | 23.1  | 20.6  | 17.2  | 13.3  | 10.2  | 9.5   | 10.8  | 14.1  | 17.3  | 20.1  | 22.4 |
|                                                                                  | DBStd       | 5.96   | 3.50  | 3.53  | 3.01  | 2.98  | 2.67  | 2.18  | 2.38  | 2.77  | 3.52  | 3.84  | 3.86  | 3.64 |
|                                                                                  | HDD10.0     | 96     | 0     | 0     | 0     | 1     | 6     | 24    | 38    | 23    | 4     | 1     | 0     | 0    |
|                                                                                  | HDD18.3     | 1212   | 2     | 3     | 12    | 56    | 158   | 245   | 273   | 232   | 134   | 67    | 23    | 7    |
|                                                                                  | CDD10.0     | 2601   | 442   | 368   | 328   | 216   | 107   | 29    | 23    | 49    | 127   | 227   | 302   | 383  |
|                                                                                  | CDD18.3     | 676    | 185   | 137   | 82    | 21    | 1     | 0     | 0     | 0     | 8     | 34    | 75    | 133  |
|                                                                                  | CDH23.3     | 6412   | 1906  | 1338  | 595   | 101   | 1     | 0     | 0     | 1     | 50    | 283   | 761   | 1377 |
|                                                                                  | CDH26.7     | 2535   | 862   | 569   | 162   | 11    | 0     | 0     | 0     | 0     | 6     | 60    | 288   | 578  |
| Wind                                                                             | WSAvg       | 4.3    | 4.6   | 4.4   | 4.1   | 3.7   | 3.8   | 4.0   | 4.1   | 4.4   | 4.5   | 4.5   | 4.6   | 4.7  |
| Precipitation                                                                    | PrecAvg     | 715    | 82    | 74    | 52    | 45    | 42    | 63    | 53    | 44    | 52    | 51    | 74    | 84   |
|                                                                                  | PrecMax     | 1196   | 225   | 156   | 157   | 236   | 110   | 155   | 214   | 119   | 207   | 145   | 296   | 236  |
|                                                                                  | PrecMin     | 372    | 0     | 14    | 0     | 1     | 0     | 20    | 3     | 3     | 3     | 0     | 7     | 15   |
|                                                                                  | PrecStd     | 190    | 59    | 41    | 41    | 53    | 33    | 36    | 45    | 34    | 43    | 37    | 63    | 58   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 37.2  | 37.4  | 32.2  | 28.2  | 22.5  | 17.7  | 18.7  | 21.6  | 27.5  | 30.7  | 35.9  | 36.2 |
|                                                                                  |             | MCWB   | 19.6  | 19.6  | 18.1  | 16.1  | 13.6  | 11.6  | 11.7  | 11.8  | 13.2  | 15.2  | 17.6  | 17.4 |
|                                                                                  | 2%          | DB     | 35.0  | 34.5  | 30.4  | 25.9  | 20.7  | 16.5  | 16.6  | 19.4  | 24.7  | 28.3  | 32.4  | 33.9 |
|                                                                                  |             | MCWB   | 19.0  | 19.4  | 17.6  | 15.9  | 13.1  | 11.1  | 10.1  | 10.8  | 12.8  | 14.8  | 16.7  | 17.5 |
|                                                                                  | 5%          | DB     | 33.3  | 32.4  | 28.5  | 24.3  | 19.4  | 15.4  | 15.2  | 17.7  | 22.4  | 26.5  | 29.8  | 31.9 |
|                                                                                  |             | MCWB   | 18.9  | 18.5  | 17.0  | 15.1  | 12.4  | 10.5  | 9.9   | 10.2  | 12.1  | 14.5  | 16.2  | 17.0 |
|                                                                                  | 10%         | DB     | 31.3  | 30.2  | 26.6  | 22.7  | 18.1  | 14.1  | 13.8  | 16.1  | 20.5  | 24.6  | 27.7  | 30.1 |
|                                                                                  |             | MCWB   | 18.7  | 18.0  | 17.1  | 14.6  | 11.7  | 10.1  | 9.3   | 9.3   | 11.5  | 14.0  | 15.9  | 16.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.5  | 22.8  | 21.0  | 18.9  | 16.0  | 13.6  | 14.6  | 14.7  | 16.1  | 18.2  | 20.3  | 21.1 |
|                                                                                  |             | MCDB   | 30.2  | 30.9  | 25.8  | 23.4  | 18.2  | 15.0  | 15.9  | 18.8  | 22.1  | 23.6  | 28.3  | 28.1 |
|                                                                                  | 2%          | WB     | 21.5  | 21.4  | 20.2  | 17.4  | 14.8  | 12.7  | 12.6  | 12.7  | 14.7  | 17.1  | 19.1  | 20.1 |
|                                                                                  |             | MCDB   | 28.8  | 28.7  | 24.6  | 22.2  | 18.3  | 14.4  | 13.8  | 16.0  | 20.4  | 23.0  | 25.8  | 26.3 |
|                                                                                  | 5%          | WB     | 20.8  | 20.4  | 19.2  | 16.3  | 13.6  | 11.8  | 11.1  | 11.4  | 13.6  | 16.2  | 18.3  | 19.4 |
|                                                                                  |             | MCDB   | 27.9  | 27.5  | 24.2  | 21.9  | 17.5  | 13.9  | 13.4  | 15.8  | 19.1  | 22.5  | 24.7  | 25.9 |
|                                                                                  | 10%         | WB     | 20.1  | 19.5  | 18.4  | 15.5  | 12.7  | 11.0  | 9.9   | 10.2  | 12.5  | 15.3  | 17.5  | 18.7 |
|                                                                                  |             | MCDB   | 27.4  | 26.7  | 23.9  | 21.0  | 16.8  | 13.5  | 13.2  | 15.0  | 18.3  | 22.3  | 24.0  | 25.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.5  | 11.4  | 10.3  | 10.1  | 8.6   | 7.4   | 7.7   | 8.9   | 10.0  | 11.2  | 11.7  | 12.0 |
|                                                                                  |             | MCDBR  | 13.4  | 13.4  | 12.2  | 11.2  | 9.7   | 8.9   | 9.3   | 10.5  | 11.8  | 12.8  | 14.2  | 14.0 |
|                                                                                  |             | MCWBR  | 4.5   | 4.6   | 4.9   | 4.8   | 4.6   | 4.7   | 5.1   | 5.4   | 5.5   | 5.5   | 5.2   | 5.0  |
|                                                                                  | 5% WB       | MCDBR  | 11.3  | 11.5  | 10.0  | 10.0  | 8.1   | 6.9   | 7.0   | 9.0   | 9.7   | 11.3  | 11.0  | 11.2 |
|                                                                                  |             | MCWBR  | 4.8   | 4.7   | 5.0   | 4.9   | 4.5   | 4.2   | 4.9   | 5.7   | 5.6   | 6.0   | 5.1   | 5.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.343  | 0.341 | 0.314 | 0.296 | 0.280 | 0.275 | 0.270 | 0.278 | 0.309 | 0.323 | 0.343 | 0.335 |      |
|                                                                                  | taud        | 2.535  | 2.560 | 2.637 | 2.647 | 2.642 | 2.660 | 2.659 | 2.617 | 2.511 | 2.507 | 2.492 | 2.545 |      |
|                                                                                  | Ebn at Noon | 996    | 979   | 975   | 943   | 913   | 894   | 916   | 949   | 964   | 986   | 989   | 1007  |      |
|                                                                                  | Edh at Noon | 111    | 104   | 91    | 80    | 72    | 67    | 70    | 81    | 101   | 109   | 115   | 110   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.42   | 6.70  | 5.64  | 4.59  | 3.52  | 2.79  | 3.19  | 4.13  | 5.43  | 6.44  | 6.96  | 7.53  |      |
|                                                                                  | RadStd      | 0.43   | 0.51  | 0.41  | 0.28  | 0.21  | 0.21  | 0.19  | 0.25  | 0.34  | 0.51  | 0.72  | 0.50  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (13 neighbors) | +0.58 | N/A     | N/A    | +1.26   | N/A   | N/A   | -18         | -84     | +212    | +145    |

Nomenclature: See separate page